

PHYS 320 Examination

Thermodynamics and Statistical Mechanics

Instructions:

- Answer all questions.
 - For Questions 1–4, choose the best option.
 - For Questions 5–8, write brief but precise answers.
1. Which thermodynamic potential is minimized at equilibrium for a system held at constant temperature and pressure?

(A) Internal energy U
(B) Helmholtz free energy F
(C) Gibbs free energy G
(D) Entropy S
 2. For a reversible adiabatic process involving an ideal gas, which relationship holds?

(A) $PV = \text{constant}$
(B) $TV^{\gamma-1} = \text{constant}$
(C) $P^{1-\gamma}T^{\gamma} = \text{constant}$
(D) $VT = \text{constant}$
 3. What is the physical interpretation of the partition function Z in statistical mechanics?

(A) The average energy of the system
(B) The number of accessible microstates at a given energy
(C) A sum over Boltzmann factors for all states, encoding thermodynamic information
(D) The entropy of the system divided by Boltzmann's constant
 4. The Carnot efficiency for a heat engine operating between hot reservoir at temperature T_H and cold reservoir at T_C is given by:

(A) $\eta = 1 - \frac{T_C}{T_H}$
(B) $\eta = \frac{T_H - T_C}{T_C}$
(C) $\eta = \frac{T_H}{T_C}$

(D) $\eta = 1 - \frac{T_H}{T_C}$

5. State all three laws of thermodynamics and explain their physical significance. How do these laws constrain the behavior of physical systems?
6. Derive the relationship between entropy and the number of microstates using Boltzmann's formula $S = k_B \ln \Omega$. What does this tell us about the directionality of spontaneous processes?
7. Explain the difference between isothermal and adiabatic processes for an ideal gas. For each, describe what happens to temperature, internal energy, and entropy during expansion.
8. Discuss the concept of chemical potential and its role in phase equilibrium. How does the equality of chemical potentials across phases determine equilibrium conditions, such as in vapor-liquid coexistence?